

## SELEKNAS 2024

### Modul: Mobile (Android)

#### Software Requirement

IDE : Android Studio

Bahasa Programming : Java or Kotlin

Target Android : SDK API 33



Esemka Library wants to develop business online. Esemka Library asks you to develop a mobile application (android). Application helps customer to buy through online. Esemka Library already have Backend API, so you must connect the mobile application with Backend API. All endpoint on Backend API must using authentication except Login endpoint. Backend API using bearer token as authentication. You can access backend API with run the "EsemkaLibrary.exe", after that you can read the backend API documentation with browser with URL is "<http://localhost:5000/swagger>". The backend API documentation has already grouped by the screen.

Color Palate:

#EEE3CB

#D7C0AE

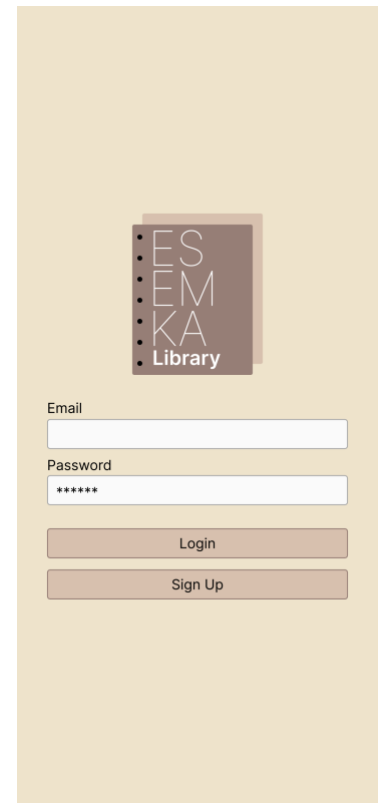
#967E76

## 1. Login

On Login screen have Email Textfield and Password Textfield. When user tap Login button, the application will validate based on Backend API. If Email and Password is valid, then the application will show Home Screen. If invalid, then application will show error message "Email or Password is invalid".

The authentication using Bearer token with token lifetime is 10 minutes. After token expired, the application will request new token.

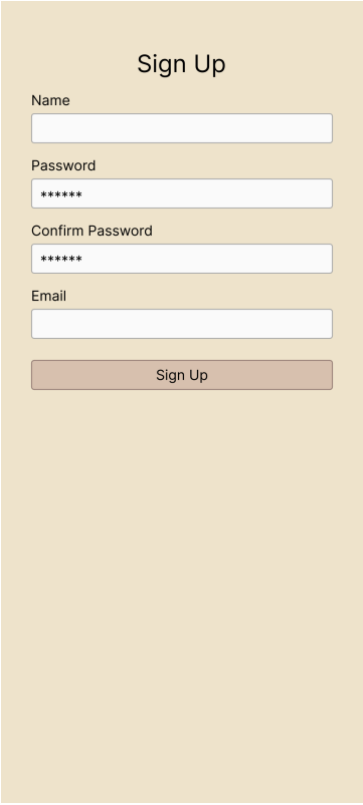
| Email                        | Password     |
|------------------------------|--------------|
| iclarricoates3@clickbank.net | fTa9al71rEm  |
| rlochran4@hatena.ne.jp       | YLhGt2JDH01N |
| ebruniges5@devhub.com        | 5lQ233NlzLc  |
| ttolotti6@dropbox.com        | QHTXNz       |
| tbrockie7@google.ruu         | idBLZ9       |



The login screen features a light beige background. At the top center is a logo consisting of a dark brown square with the text 'ES', 'EM', and 'KA' stacked vertically in white, and 'Library' in a smaller font below it. Below the logo are two input fields: 'Email' and 'Password'. The 'Email' field is a simple white rectangle with a thin border. The 'Password' field is a white rectangle with a thin border and contains six asterisks. Below these fields are two buttons: 'Login' and 'Sign Up'. Both buttons are dark brown with white text.

## 2. Sign Up

On Sign Up screen, the user must be filling the form. After filled the form and user tap on Sign Up button, the application will validate the Name must be filled, the Password must contain at least 8 characters, one uppercase, one lowercase, and one number, the Confirm Password must same with Password, and the Email must correct format. If all data has valid, the application will save the data on Backend API and return to Login Screen.

A UI mockup of a 'Sign Up' form. The form is titled 'Sign Up' at the top. It contains four input fields: 'Name', 'Password', 'Confirm Password', and 'Email'. Each field has a label above it and a corresponding input box. The 'Password' and 'Confirm Password' fields show asterisks (\*\*\*\*\*). Below the input fields is a 'Sign Up' button. The entire form is set against a light beige background.

Sign Up

Name

Password

Confirm Password

Email

Sign Up

### 3. Home

On the **Home** screen, all books are displayed in a grid format, and there are four menus at the bottom of the screen: **Home**, **Forum**, **My Cart**, and **My Profile**. When the user types in the **Search** field, the application filters the book data based on the entered search text. If the user taps on a book view, the application will navigate to the **Book Detail** screen.

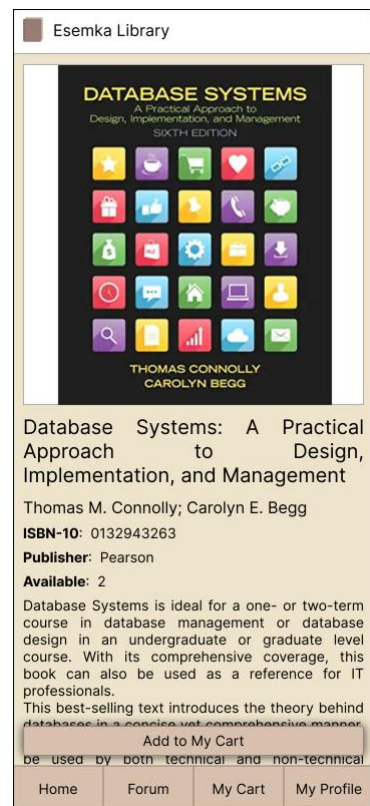
- When the **Home** menu is tapped, the application displays the Home screen.
- When the **Forum** menu is tapped, the application navigates to the **Forum** screen.
- When the **My Cart** menu is tapped, the application displays the **Cart** screen.
- When the **My Profile** menu is tapped, the application shows the **Profile** screen.



#### 4. Book Detail

On Book Detail screen, will show book Picture, Title/Name, Authors, ISBN, Publisher, Available, and Description.

When user tapped on Add to My Cart button, the application will add the book to My Cart and go to My Cart. Validate the My Cart must be fill with the unique books.



## 5. My Cart

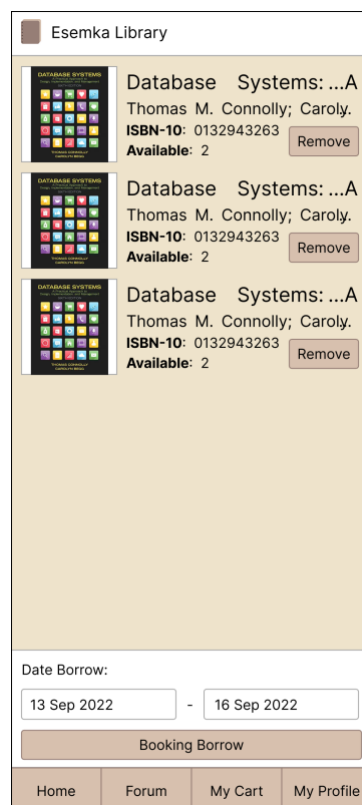
On **My Cart** screen will show selected books with their details, a **start date** for borrowing and an **end date**, and **Booking Borrow Button**. The **end date** must be disable.

When the user taps the remove button, the corresponding book will be removed from My Cart.

Tapping on the **Start Date** field will open a date picker dialog. Once the user selects a **start date**, the **end date** will automatically be set to three days after the start date. The end date field will be disabled to prevent changes.

When the user taps the **Booking Borrow button**, the borrowing details will be saved to the backend API, and the **My Cart** will be cleared.

If the application is closed and reopened, the user's data (books in "My Cart") will remain saved and displayed. (Persistent Data)



## 6. My Profile

On My Profile screen, the application will show the Photo Picture, Name, Email, and My Borrowing History.

When user tap on Upload Photo button, user must select the photo, and the app will upload select photo to backend.

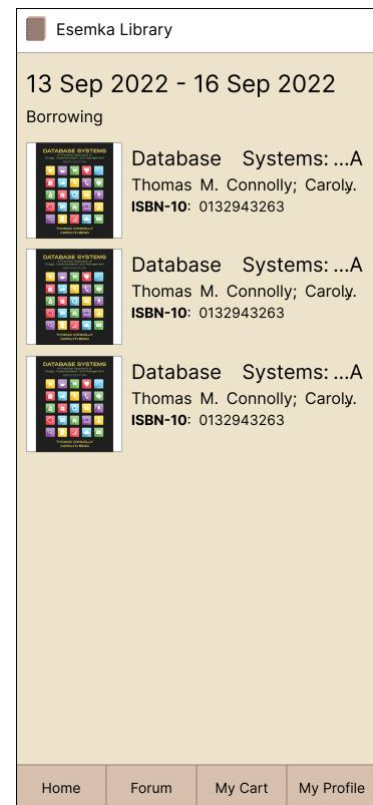
When User tap on Logout button, the app will delete the session and will show the Login screen.

When User tap on My Borrowing History, the application will go to Borrowing Detail.



## 7. Borrowing Detail

On View Transaction, will show transaction start and end date borrowing, status, and list of books with the details.

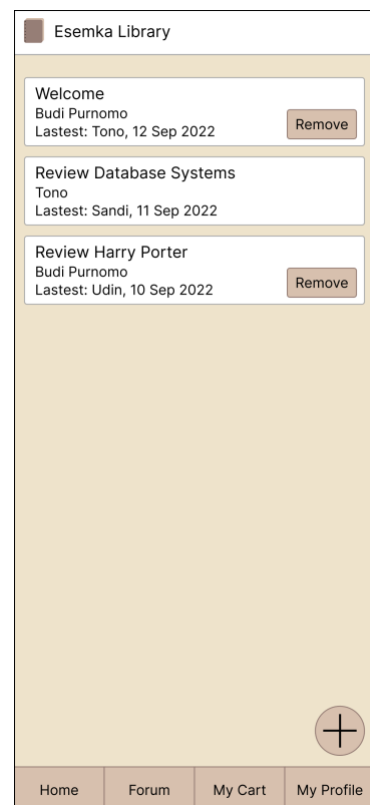




## 8. Forum

The Forum screen displays all available discussion threads, showing the **thread title**, the **creator's name**, and the **date of the latest post**. Only the thread creator could delete their thread. When the creator taps the "Remove" button, a confirmation box appears, asking whether they are sure about deleting the thread. If the user confirms by choosing "Yes," the thread will be deleted from the backend API.

When the user taps the "Add" button, the application navigates to a screen for adding a new thread. Additionally, tapping on any thread will open a detailed view of that thread, where the user can see all associated posts and interactions.



## 9. Add Thread

On Add Thread screen, will show form (Subject and Body). When user tap on Add Thread Button, the application will validate. If all data is valid, then the application will save threat on Backend API and return to Forum screen.

The screenshot shows the 'Add Thread' screen of the 'Esemka Library' application. The screen features a title bar at the top with the text 'Esemka Library' and a back arrow icon. Below the title bar, there are two input fields: 'Subject' and 'Body'. The 'Subject' field is a single-line text input, and the 'Body' field is a multi-line text area. Below these fields, there is a button labeled 'Add Thread'. At the bottom of the screen, there is a navigation bar with four tabs: 'Home', 'Forum', 'My Cart', and 'My Profile'.

## 10. Thread Detail

On Thread detail screen, will show the thread and all messages. Only creator message can remove the message.

When remove button tapped, the confirmation dialog will be show. If the user chooses Yes, then the message will be deleted.

When user tap on Add Reply button, the message will save on Backend API.

